

W.O.P.R. / the open-top revision

Side-by-side LED banks. Rear-mounted electronics. Removable pen cups. Gray and flush white text.

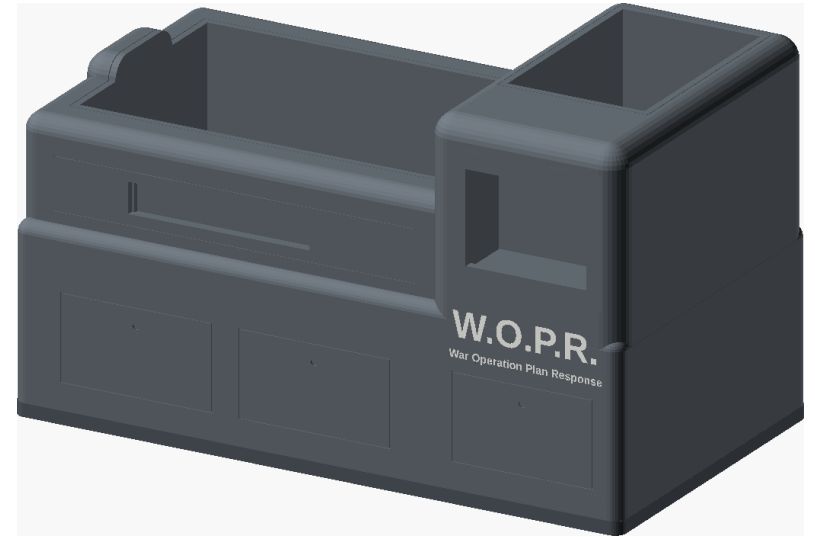


279.4 x 155 x 165 mm | 11.00 x 6.10 x 6.50 in | 3 mm nominal shell

OpenSCAD model render. Colored LEDs and display are hardware previews, not printed material.

Continuous banks, hidden fasteners

Eight boards total: four adjacent boards behind one long window on each side.



CASE WITHOUT ELECTRONICS

The open top contains two removable pen wells, isolated from the electronics.

Both long sides carry white W.O.P.R. / War Operation Plan Response lettering.

The main touchscreen is on one side, directly above its logo.

Each row spans 84.72 mm; only 0.30 mm separates neighboring PCB edges.

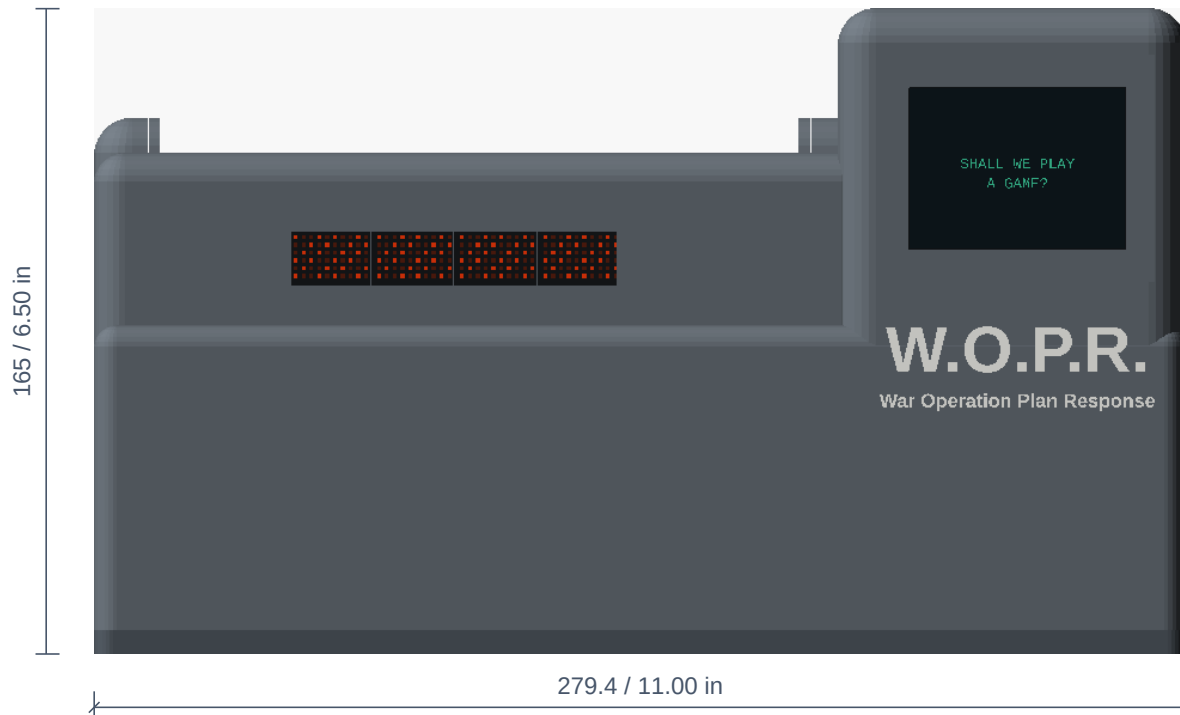
Every board loads from behind its bezel. No mounting screws are exposed on the front.

Remove the pen cups and base for rear access. Fit the electronics first, then lower the cups into place.

Overall envelope and panel placement

All views below come from the same SCAD geometry. Dimensions are nominal, not a film-prop measurement.

PRIMARY LONG SIDE



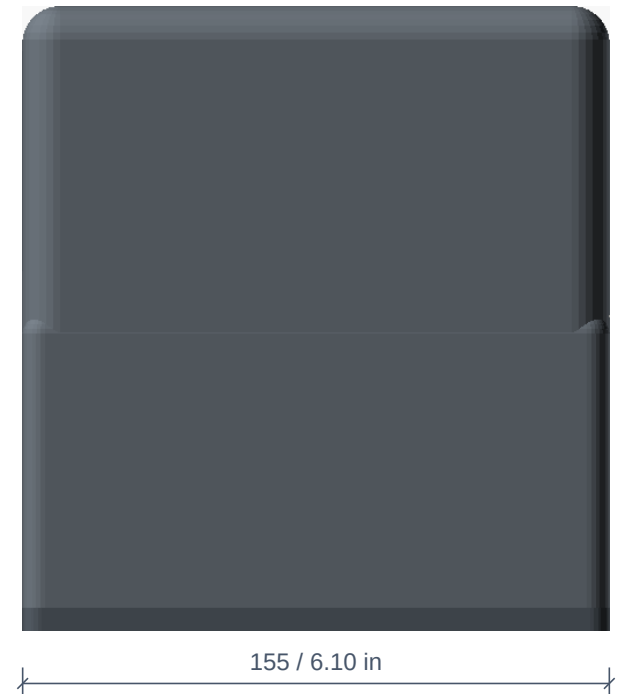
LED centers: X = 60.118, 81.373, 102.628, 123.883.

TFT center: X = 236; Z = 124. Lower case top: Z = 128.

Spine top: Z = 137. Tower top: Z = 165.

Case splits at X = 140 for a smaller printer.

TOWER END



Rear TFT pocket: 65.6 x 53.6.

Front TFT window: 55.6 x 41.6.

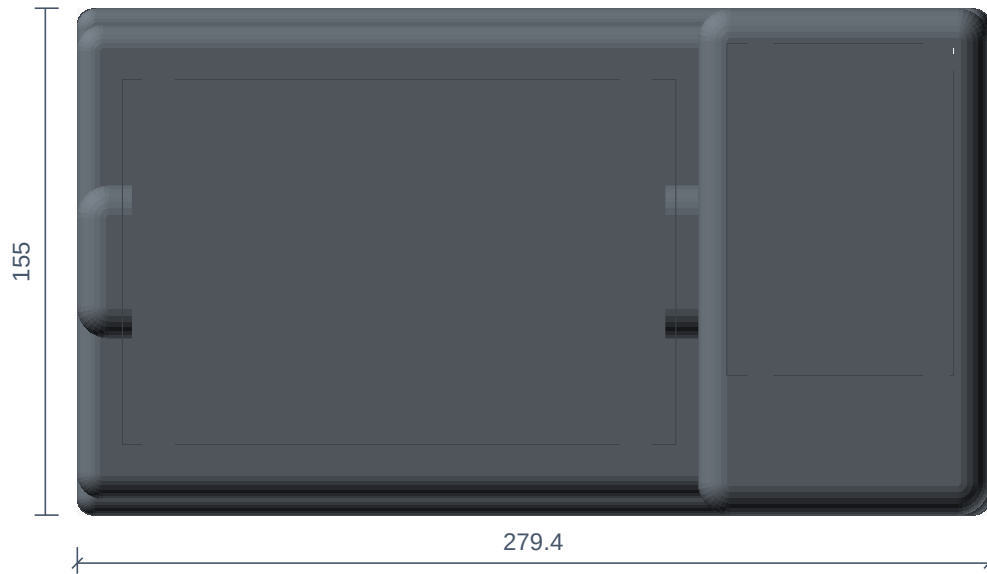
Rear LED bank pocket:

85.32 wide x 18.38 high.

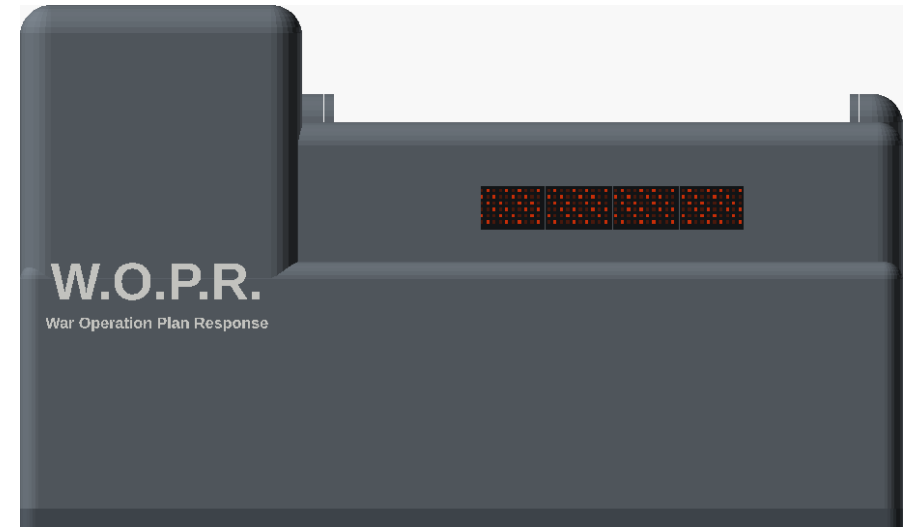
Top, rear and service access

Common axes: X along the 11-inch length, Y front to rear, Z up from the base.

TOP



OPPOSITE LONG SIDE



BOTTOM / TWO REMOVABLE HALVES

Main pen well: 163 x 105 inside; floor at Z = 43.

Tower pen well: 63 x 95 inside; floor at Z = 68.

Both cups have 3 mm walls and floors.

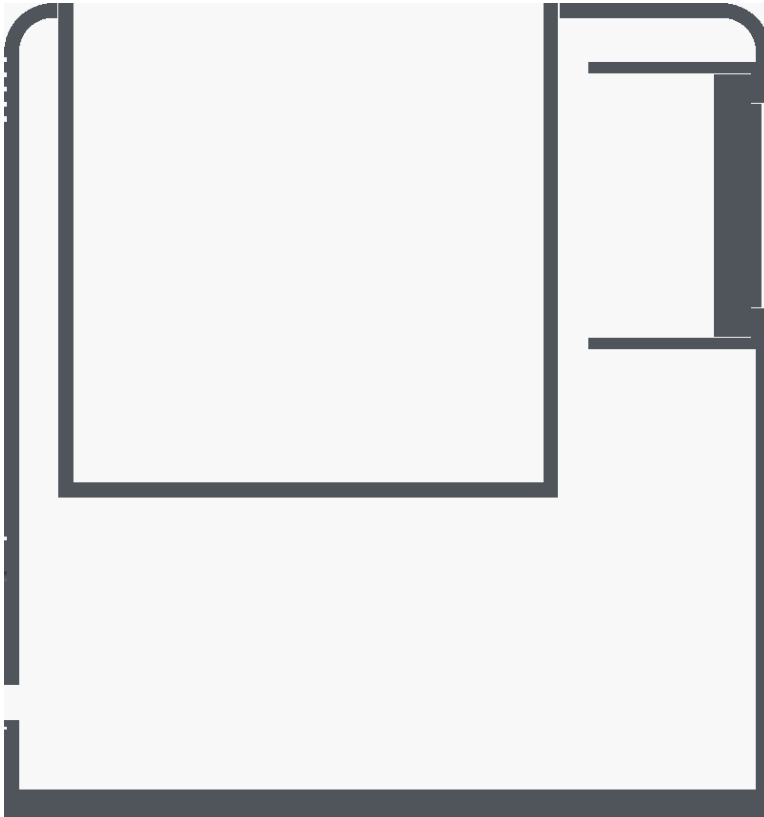
Lift the cups out to reach the rear fasteners.

Eight M3 base screws; 3.3 clearance / 2.5 pilots.

Cup-to-case clearance: 0.30 mm per side.

A full inch behind the display

Section through X = 236. The bay is open at the rear for wires and installation from below.



SCAD SECTION / LOOKING ALONG X

DEPTH CHAIN / DETAIL



Rear-loading pocket: 65.6 x 53.6; front bezel hides PCB.

Four hidden blind pilots: diameter 1.8 for M2 screws.

V2 hole pitch: 59.690 x 47.498 (landscape).

Front glass is nominally 2 mm below the case face.

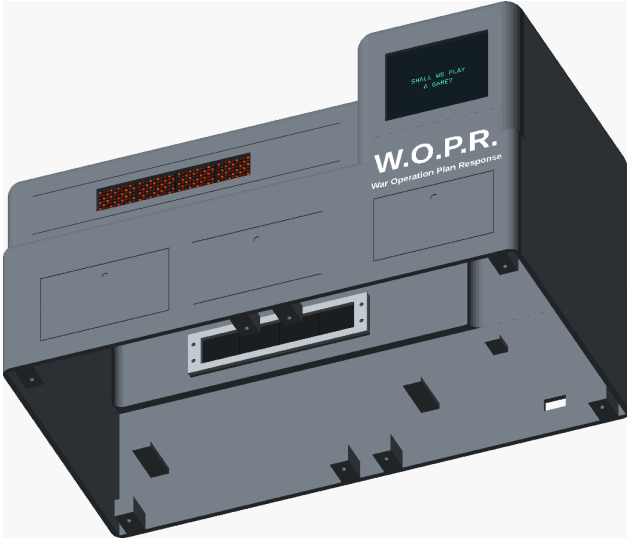
Tower cup starts at Y = 43, beyond the 36.9 mm bay.

The 25.4 mm bay excludes the 9.5 mm hardware depth.

Mount positions follow manufacturer CAD; assembly thickness and tolerances still need a physical check.

Install from the inside

Rear access shown with the base and removable pen cups removed. Physical fit remains untested.



PRINT FILES

right-shell-two-color.3mf: one object, gray + white parts.

gray-left.stl / gray-right.stl: split structural case.

white-right.stl: aligned lettering; import with gray-right.

base-left.stl / base-right.stl: removable underside.

cup-left / cup-right / cup-tower.stl: removable pen cups.

led-retainer.stl: print two; screws enter from inside.

Nominal shell: 3 mm. White text: 0.8 mm deep, flush.

splice.stl: print six. Test fit-coupon.stl before the case.

Start at 0.2 mm layers. Use a bed at least 180 x 180 mm.

Assign the two filament colors explicitly in the slicer.

UNDERSIDE / REAR MOUNTING ACCESS

CHECKED: Eleven STL exports are watertight. Structural parts are connected solids.

CHECKED: Rear insertion, cup removal, open pen wells and display clearance pass geometry probes.

NOT CHECKED: Actual print, hardware fit, touch access, electrical power or operating temperature.

Proportions are adapted from visual references. This is a review draft, not a measured replica.

SOURCES / MANUFACTURER DATA AND VISUAL REFERENCE

Seed product + board CAD	https://wiki.seeedstudio.com/rgb_matrix_for_xiao/
Adafruit #3315 / 65 x 53 x 9.5 mm	https://www.adafruit.com/product/3315
Adafruit V2 mounting geometry	https://github.com/adafruit/Adafruit-2.4-TFT-FeatherWing-PCB
Miniatua / visual proportions only	https://www.miniatua.com/work/wopr/